

8.3 Trigonometric integrals:-

-> Product of powers of sines and cosines
we begin with integrals of the form

$$\int \sin^m x \cos^n x \, dx.$$

case 1 if m is odd, we write m as $2k+1$ and use the identity $\sin^2 x = 1 - \cos^2 x$ to obtain.

as in example no 1

ex, evaluate

$$\int \sin^3 x \cos^2 x \, dx.$$

This is the example of case 1.

$$\int \sin^3 x \cos^2 x \, dx = \int \sin^2 x \cos^2 x \sin x \, dx$$

= we know that $\sin^2 x + \cos^2 x = 1$

$$\sin^2 x = 1 - \cos^2 x.$$

is odd

$$\sin x \, dx = -d \cos x.$$

let $u = \cos x$.

$$\frac{du}{dx} = -\sin x$$

$$du = -\sin x \, dx.$$

$$-du = \sin x \, dx.$$

$$\int (1 - \cos^2 x) (\cos^2 x) (-d \cos x)$$

$$\int (1 - u^2) (u^2) (-du)$$

$$\int (u^2 - u^4) (-du)$$

$$\int (u^4 - u^2) (du)$$

$$\frac{u^5}{5} - \frac{u^3}{3} + c = \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + c$$

ex2

Evaluate.

$$\int \cos^n x \, dx$$

so This is an example of case 2, where $m=0$ is even and $n=5$ is odd.

$$\int \cos^5 x \, dx = \int \cos^4 x \cos x \, dx = \int (1 - \sin^2 x)^2 d(\sin x)$$

$$= \int (1 - u^2)^2 du$$

$$= \int (1 - 2u^2 + u^4) du$$

$$= \int (1 - 2u^2 + u^4) du$$

$$= u - \frac{2}{3} u^3 + \frac{1}{5} u^5 + C = \sin x - \frac{2}{3} \sin^3 x + \frac{1}{5} \sin^5 x + C$$

$$\sin^5 x + C$$

$$\text{let } u = \cos x$$

$$\frac{du}{dx} = -\sin x$$

$$du = -\sin x \, dx$$

$$\cos x \, dx = d(\sin x)$$

$$u = \sin x$$

ex 2

$$\int \cos n \, dn$$

$$\cos n \, dn \frac{dn}{dn} = \sin n$$

$$\int \cos^4 n \cdot \cos n \, dn$$

$$\text{let } \frac{du}{dn} = \sin n$$

$$\int (1 - \sin^2 n)^2 \cdot du$$

$$du = \cos n \, dn$$

$$\int (1 - u^2)^2 \cdot du$$

$$\text{let } u = \sin n$$

$$\int 1^2 - 2u^2 + u^4 \, du$$

$$u - \frac{2}{3} u^3 + \frac{u^5}{5} + C \quad \text{Ans.}$$

case 3

if both m and n are even in

$\int \sin^m x \cos^n x dx$, we substitute

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

example no 3

$$\int \sin^2 x \cos^4 x dx$$

solution:-

This is an example of case 3

$$\int \left(\frac{1 - \cos 2x}{2} \right) \cdot \left(\frac{1 + \cos 2x}{2} \right)^2 dx$$

$$\frac{1}{8} = \int (1 - \cos 2x) (1 + 2\cos 2x + \cos^2 2x) dx$$

$$\frac{1}{8} \int (1 + \cos 2x - \cos^2 2x - \cos^3 2x) dx$$

$$\frac{1}{8} \left[x + \frac{1}{2} \sin 2x - \int (\cos^2 2x - \cos^3 2x) dx \right]$$

$$\text{For } \int \cos^2 2x = \frac{1 + \cos 4x}{2}$$

$$\frac{1}{2} \int (1 + \cos 4x)$$

$$\frac{1}{2} \left[x + \frac{1}{4} \sin 4x \right]$$

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8 Now for $\cos^3 x$, we use

$$\int \cos^3 x \, dx = \int (1 - \sin^2 x) \cos x \, dx$$
$$= \frac{1}{2} \int (1 - u^2) \, du = \frac{1}{2} \left(\sin x - \frac{1}{3} \sin^3 x \right)$$

Combining everything and simplifying, we get

$$\int \sin^3 x \cos x \, dx = \frac{1}{16} \left(x - \frac{1}{4} \sin x + \frac{1}{3} \sin^3 x \right) + C$$